LN6 Stochastic Differential Equations

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Outlines

Textbook: Arbitrage Theory in Continuous Time: Chapter 5

1. Stochastic Differential Equations
2. Geometric Brownian Motion
3. Linear SDE
4. Infinitesimal Generator
5. Partial Differential Equation

Stochastic Differential Equations

Let $M(n, d)$ denote the class of $n \times d$ matrices, and consider as given the following objects.

- W: d-dimensional Wiener process.
- μ: $\mathbb{R}^+ \times \mathbb{R}^n \to \mathbb{R}^n$
- μ: $\mathbb{R}^+ \times \mathbb{R}^n \to M(n, d)$
- $X_0 \in \mathbb{R}^n$

Stochastic Differential Equations

The stochastic differential equation (SDE):

$$\begin{aligned} dX_t &= \mu(t, X_t)dt + \sigma(t, X_t)dW_t \\ X_0 &= x_0 \end{aligned} \tag{1}$$

We want to find a process X satisfying the integral equation

$$X_t = x_0 + \int_0^t \mu(s, X_s)ds + \int_0^t \sigma(s, X_s)dW_s, \text{ for all } t \geq 0. \tag{2}$$

The standard method for proving the existence of a solution to the SDE above is to construct an iteration scheme of Cauchy–Picard type:

$$\begin{aligned} X_t^0 &= x_0 \\ X_t^{n+1} &= x_0 + \int_0^t \mu(s, X_s^n)ds + \int_0^t \sigma(s, X_s^n)dW_s \end{aligned} \tag{3}$$

Does it converge?

Stochastic Differential Equations

Proposition 1

Suppose that there exists a constant K s.t.

$$\begin{aligned} \|\mu(t,x) - \mu(t,y)\| &\leq K\|x-y\| \\ \|\sigma(t,x) - \sigma(t,y)\| &\leq K\|x-y\| \\ \|\mu(t,x)\| + \|\sigma(t,x)\| &\leq K(1+\|x\|) \end{aligned} \tag{4}$$

Then there exists a unique solution to the SDE (1). Furthermore,

1. *X is $\mathcal{F}_t^W$-adapted.*
2. *X has continuous paths.*
3. *X is a Markov process.*
4. *There exists a constant C s.t. $E[\|X_t\|^2] \leq Ce^{Ct}(1+\|x_0\|^2)$.*

Geometric Brownian Motion

Geometric Brownian motion:

$$\begin{cases} dX_t = \alpha X_t dt + \sigma X_t dW_t \\ X_0 = x_0 \end{cases} \tag{5}$$

Proposition 2

The solution to the equation (5) is given by

$$X_t = x_0 \cdot \exp\left\{\left(\alpha - \frac{\sigma^2}{2}\right) t + \sigma W_t\right\} \tag{6}$$

The expected value is given by

$$E[X_t] = x_0 e^{\alpha t} \tag{7}$$

Proof of (6) is shown on next page.

Proof of (6)

Let $Z_t = \ln X_t$, then

$$\begin{aligned} dZ_t &= \frac{1}{X_t} dX_t - \frac{1}{2}\frac{1}{X_t^2} d[X,X]_t \\ &= \frac{1}{X_t}(\alpha X_t dt + \sigma X_t dW_t) - \frac{1}{2X_t^2}\sigma^2 X_t^2 dt \\ &= \left(\alpha - \frac{\sigma^2}{2}\right) dt + \sigma dW_t \end{aligned}$$

Then

$$Z_t = \ln x_0 + \left(\alpha - \frac{\sigma^2}{2}\right) t + \sigma W_t$$

$$X_t = x_0 \cdot \exp\left\{\left(\alpha - \frac{\sigma^2}{2}\right) t + \sigma W_t\right\}$$

Stochastic Exponential

Stochastic Exponential (Doleans-Dade Exponential) Let X be a semi-martingale, $X_0 = 0$. $Z_t = 1 + \int_0^t Z_{s-} dX_s$. Then

$$Z_t = \exp\{X_t - \frac{1}{2}[X,X]_t\} \prod_{0<s\leq t} (1+\Delta X_s)\exp\{-\Delta X_s + \frac{1}{2}(\Delta X_s)^2\}$$

For continues case,

$$Z_t = \mathcal{E}(X_t) = \exp\{X_t - \frac{1}{2}[X,X]_t\}$$

The Linear SDE

Linear SDE

$$\begin{cases} dX_t = \alpha X_t dt + \sigma dW_t \\ X_0 = x_0 \end{cases} \tag{8}$$

Recall that the linear ODE

$$\frac{dx_t}{dt} = \alpha x_t + u_t \tag{9}$$

where u is a deterministic function of time, has the solution

$$x_t = e^{\alpha t} x_0 + \int_0^t e^{\alpha(t-s)} u_s ds \tag{10}$$

So, the answer may be

$$X_t = e^{\alpha t} X_0 + \sigma \int_0^t e^{\alpha(t-s)} dW_s \tag{11}$$

The Linear SDE

Proposition 3

Consider the n-dimensional linear SDE

$$\begin{cases} dX_t = (AX_t + bt)dt + \sigma_t dW_t \\ X_0 = x_0 \end{cases} \tag{12}$$

where A is an $n \times n$ matrix, b is an R^n-valued deterministic function, $\sigma \in M(n,d)$, and W an d-dimensional Wiener process. Then

$$X_t = e^{At}x_0 + \int_0^t e^{A(t-s)}b_s ds + \int_0^t e^{A(t-s)}\sigma_s dW_s \tag{13}$$

where $e^{At} = \sum_{k=0}^{\infty} \frac{A^k}{k!} t^k$.

Infinitesimal Generator

Consider the n-dimensional SDE

$$dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dW_t. \tag{14}$$

Definition 1

Given the SDE above, the **infinitesimal generator** *of X is defined, for any function $h(x) \in C^2(R^n)$, by*

$$\mathcal{A}h(x) = \sum_{i=1}^{n} \mu_i(t, x)\frac{\partial h}{\partial x_i}(x) + \frac{1}{2}\sum_{i,j=1}^{n} C_{ij}(t, x)\frac{\partial^2 h}{\partial x_i \partial x_j}(x)$$

where

$$C(t, x) = \sigma(t, x)\sigma(t, x)'$$

Itô Formula

In terms of the infinitesimal generator, the Itô formula takes the form

$$df(t, X_t) = \left\{ \frac{\partial f}{\partial t} + \mathcal{A}f \right\} dt + [\nabla_x f]\sigma dW_t$$

where

$$\nabla_x f = \left[\frac{\partial h}{\partial x_1}, \cdots, \frac{\partial h}{\partial x_n} \right].$$

Infinitesimal Generator

Markov process and I.G.

- A one parameter family $\{T(t) : t \geq 0\}$ of bounded linear operators on a Banach space L is called a semigroup if $T(0) = I$ and $T(s+t) = T(s)T(t)$ for all $s, t \geq 0$.
- I.G. of $\{T(t)\}$.

$$\mathcal{A}f = \lim_{t \to 0} \frac{1}{t}\{T(t)f - f\}$$

- Since $E[f(X(t+s))|\mathcal{F}_t^X] = T(s)f(X(t))$, then

$$\mathcal{A}f(x) = \lim_{t \to 0} \frac{E^x f(X_t) - f(x)}{t}$$

Cauchy Problem

$$\frac{\partial F}{\partial t}(t,x) + \mu(t,x)\frac{\partial F}{\partial x}(t,x) + \frac{1}{2}\sigma^2(t,x)\frac{\partial^2 F}{\partial x^2}(t,x) = 0 \qquad (15)$$
$$F(T,x) = \Phi(x)$$

We define the stochastic process X on the time interval $[t, T]$ as the solution to the SDE

$$\begin{cases} dX_s = \mu(s, X_s)ds + \sigma(s, X_s)dW_s \\ X_t = x \end{cases} \qquad (16)$$

Then

$$\mathcal{A} = \mu(t,x)\frac{\partial}{\partial x} + \frac{1}{2}\sigma^2(t,x)\frac{\partial^2}{\partial x^2}$$

Thus we may write the Cauchy problem as

$$\frac{\partial F}{\partial t}(t,x) + \mathcal{A}F(t,x) = 0 \qquad (17)$$
$$F(T,x) = \Phi(x)$$

Cauchy Problem Continued

Applying the Itô formula to the process $F(s, X_s)$ gives us

$$F(T, X_T) = F(t, x) + \int_t^T \left\{ \frac{\partial F}{\partial s}(s, X_s) + \mathcal{A}F(s, X_s) \right\} ds + \int_t^T \sigma(s, X_s) \frac{\partial F}{\partial x}(s, X_s) dW_s$$

Then

$$F(t, x) = E_{t,x}[\Phi(X_T)]$$

Feynman-Kac Stochastic Representation Formula

Proposition 4

Assume that F is a solution to

$$\frac{\partial F}{\partial t}(t,x) + \mu(t,x)\frac{\partial F}{\partial x}(t,x) + \frac{1}{2}\sigma^2(t,x)\frac{\partial^2 F}{\partial x^2}(t,x) = 0 \qquad (18)$$
$$F(T,x) = \Phi(x)$$

Assume furthermore that the process $\sigma(s,X_s)\frac{\partial F}{\partial x}(s,X_s)$ is in $\mathcal{L}^2$. Then F has the representation

$$F(t,x) = E_{t,x}[\Phi(X_T)]$$

where X satisfies the SDE

$$dX_s = \mu(s,X_s)ds + \sigma(s,X_s)dW_s \qquad (19)$$
$$X_t = x$$

Feynman-Kac Stochastic Representation Formula

Proposition 5

Assume that F is a solution to

$$\frac{\partial F}{\partial t}(t,x) + \mu(t,x)\frac{\partial F}{\partial x}(t,x) + \frac{1}{2}\sigma^2(t,x)\frac{\partial^2 F}{\partial x^2}(t,x) - rF(t,x) = 0$$
$$F(T,x) = \Phi(x) \tag{20}$$

Assume furthermore that

$$e^{-rs}\sigma(s,X_s)\frac{\partial F}{\partial x}(s,X_s)$$

is in $\mathcal{L}^2$. Then F has the representation

$$F(t,x) = e^{-r(T-t)}E_{t,x}[\Phi(X_T)]$$

Multi-Dimension

Proposition 6

Assume that $F : R_+ \times R^n \to R$ *is a solution to*

$$\begin{aligned} \frac{\partial F}{\partial t}(t,x) + \sum_{i=1}^{n} \mu_i(t,x)\frac{\partial F}{\partial x_i}(t,x) + \frac{1}{2}\sum_{i,j=1}^{n} C_{ij}(t,x)\frac{\partial^2 F}{\partial x_i \partial x_j}(t,x) \\ -rF(t,x) = 0, \\ F(T,x) = \Phi(x). \end{aligned} \tag{21}$$

Assume furthermore that the process

$$e^{-rs}\sum_{i=1}^{n} \sigma_i(s, X_s)\frac{\partial F}{\partial x_i}(s, X_s)$$

is in $\mathcal{L}^2$. *Then* F *has the representation*

$$F(t,x) = e^{-r(T-t)}E_{t,x}[\Phi(X_T)]$$

Partial Differential Equation

Proposition 7

The process $F(t, X_t)$ is a martingale with respect to the filtration $\mathcal{F}^X$

$$\iff \frac{\partial F}{\partial t} + \mathcal{A}F = 0.$$

$$\iff F(t,x) = E_{t,x}[F(T, X_T)] \quad \textit{for every } (t,x) \textit{ and } T \geq t.$$